Error Analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Error Analysis before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Nearest Neighbors

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Error Analysis trace and record the first step you can explain without help.

Use W2: Explain the limit for **Error analysis** and controlled follow-up. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Error Analysis explanation against the supplied facts before discussing it.

Use the worked case for **Error analysis** and controlled follow-up. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

> **My independent explanation, trace and limits**
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> | |
> |---|
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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Error Analysis target to a trace and a justified conclusion.

A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Error Analysis case and show the deciding evidence.

Four errors occur among five backlit photos; two among twenty ordinary photos.

Compare rates and investigate.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Of eight errors, six are on blurry images and two on clear images. Can you conclude blurry images have a higher error rate without group totals?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 18 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Error Analysis adds to Responsible Plant Classifier and keep the evidence.

Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Find the rates

> Find the rates. Apply the supplied Error Analysis facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Error Analysis", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Calculate the original error rates by lighting condition.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Explain the limit

> Explain the limit. Apply the supplied Error Analysis facts and keep your reasoning.

Does the difference prove dim light causes model errors?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Track regressions

Track regressions. Apply the supplied Error Analysis facts and keep your reasoning.

Summarize the changed results without hiding a **regression**.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Error

> Analysis facts and keep your reasoning.

Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Repair a misleading audit

> Repair a misleading audit. Apply the supplied Error Analysis facts and keep your reasoning.

A team removes all disputed records only from the new model report. Why is the comparison invalid?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Error Analysis

> Find an error cluster before choosing a controlled follow-up. Explain the labelled step before checking the changed condition.

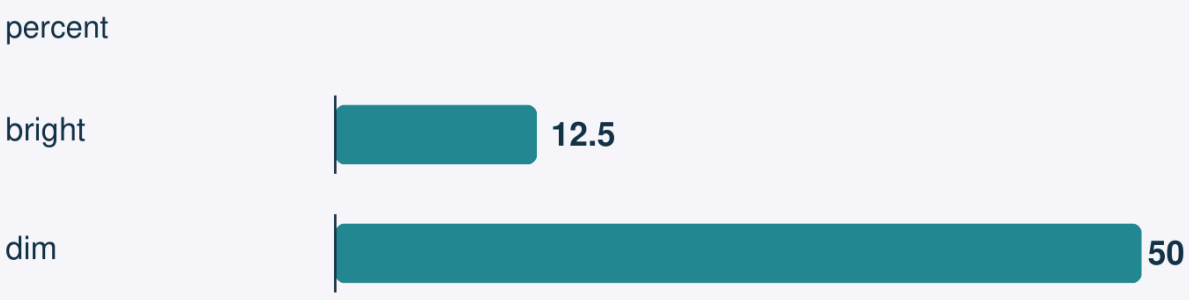
Find an error cluster before choosing a controlled follow-up.

Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/18

Errors per 100 photos in each lighting slice



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/18

Error Analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice errors

Bright: 12.5%; dim: 50%. Total errors4/14, about28.6%.





2. Inspect dim failures

Check blur, lighting and label uncertainty instead of assuming lighting caused every error.





3. Plan controlled pair

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

CHANGE ONE THING

Dim errors fall to1/6 in a later same-case replay.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Find an error cluster before choosing a controlled follow-up.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

1. Plan controlled pair

2. Compute slice errors

3. Inspect dim failures

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Plan controlled pair Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Read the picture in order.

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Dim errors fall to1/6 in a later same-case replay. Which conclusion is supported by the supplied facts?

1. The replay’s dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.

Observed improvement on reused cases is not an untouched final generalisation test.

Repair a missing building block

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

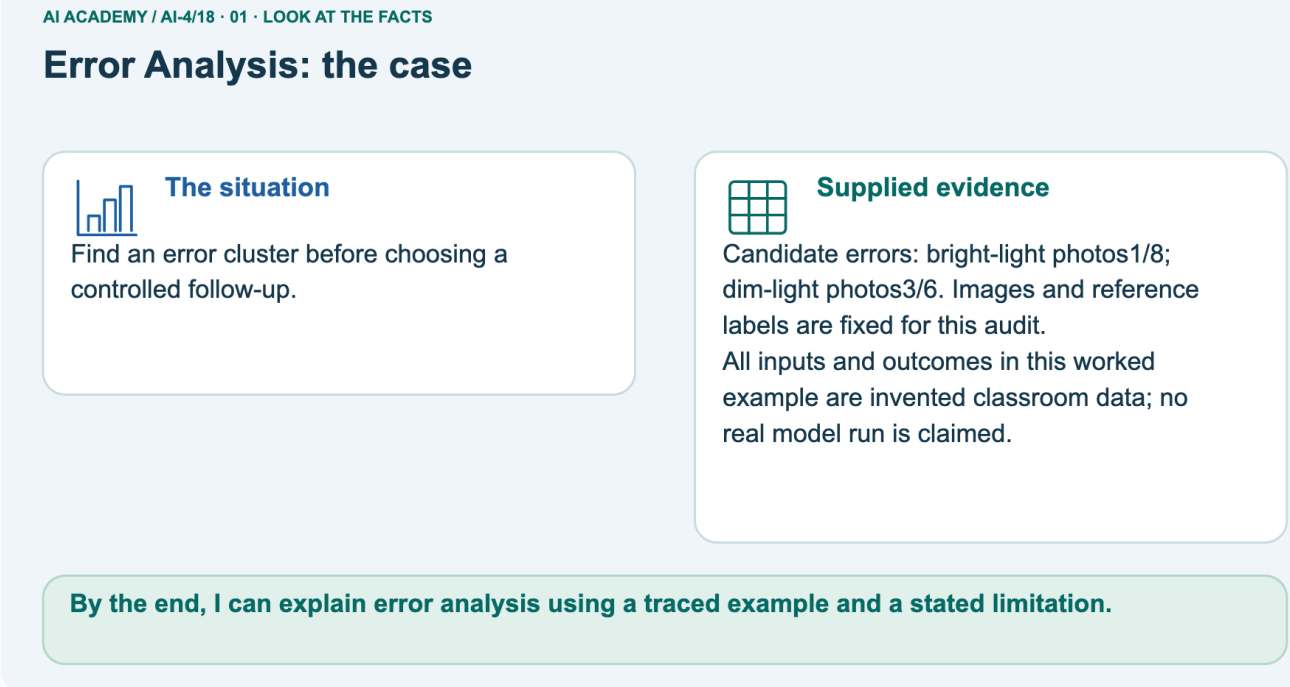
> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Error Analysis picture from its labels, then record a relevant limit.

01. Error Analysis: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

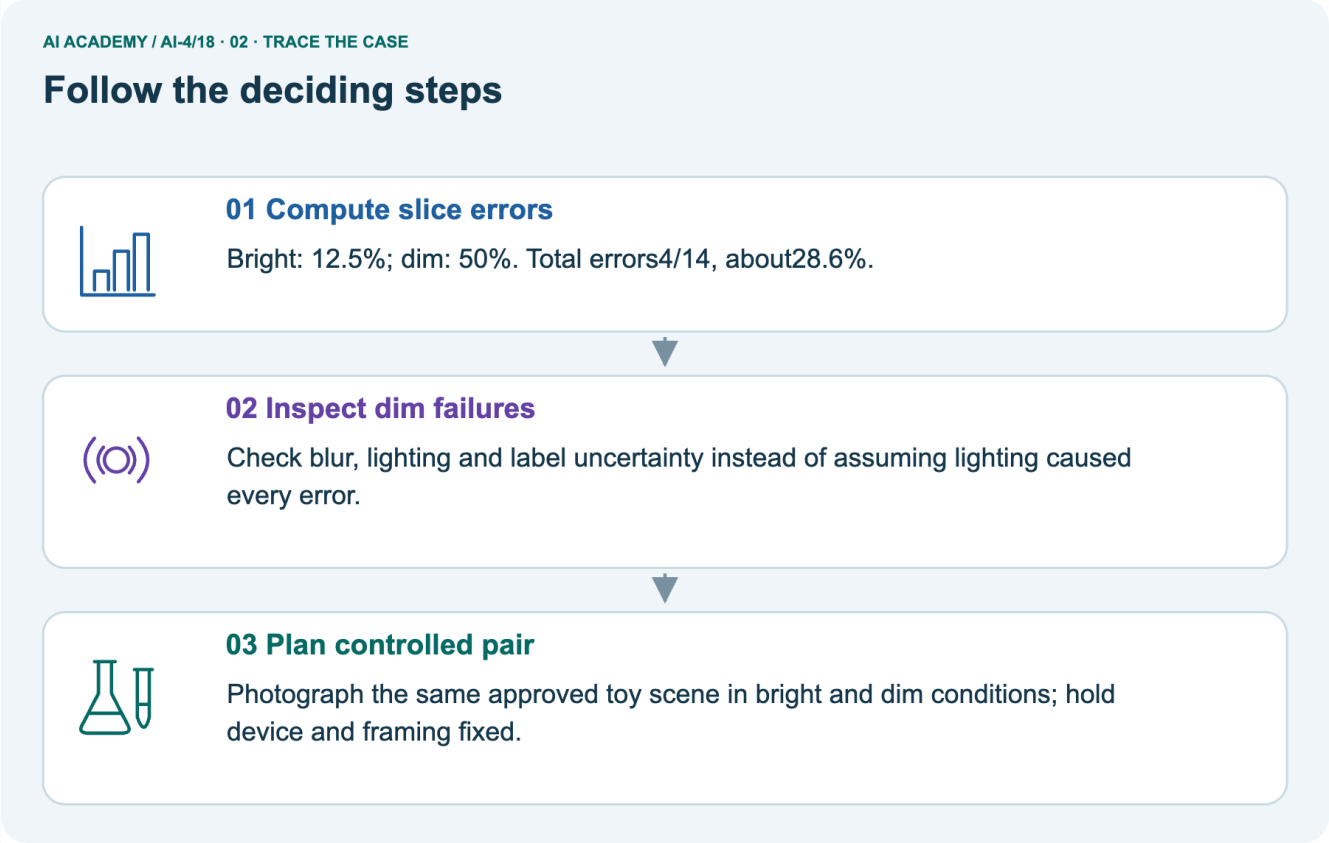
Find an error cluster before choosing a controlled follow-up. Supplied facts: Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

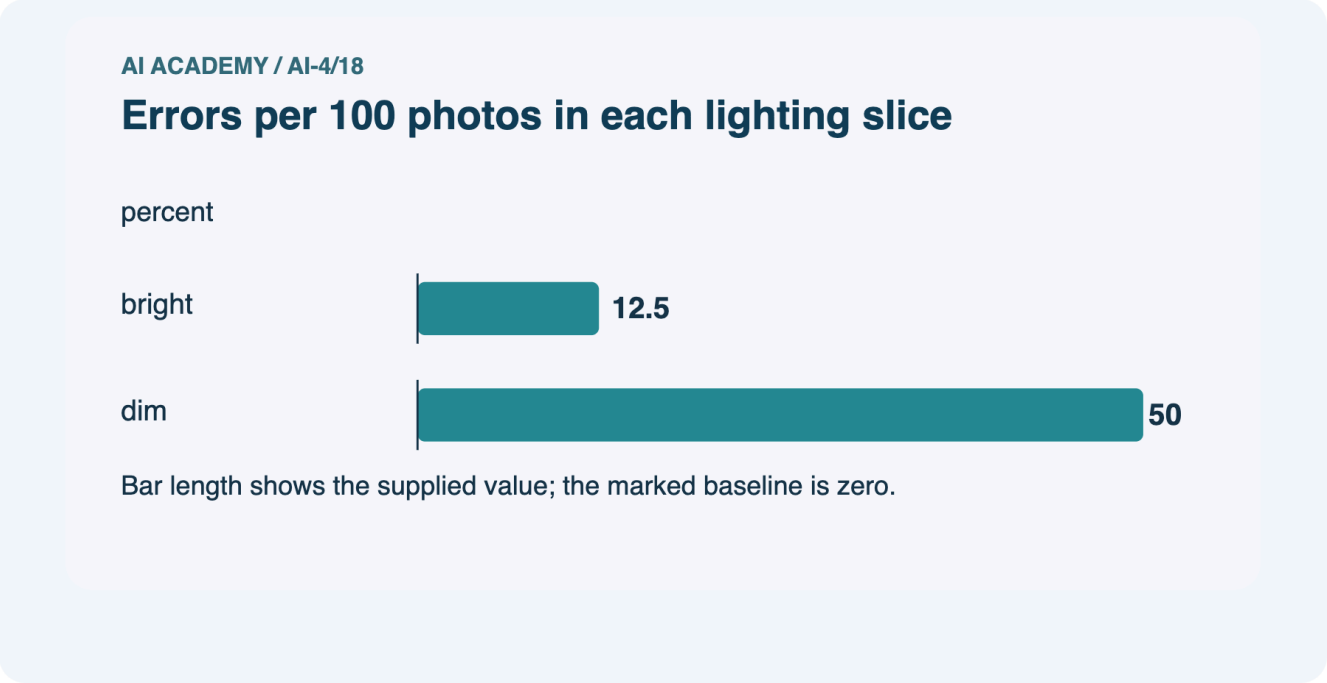
Compute slice errors: Bright: 12.5%; dim: 50%.

Total errors4/14, about28.6%. → Inspect dim failures: Check blur, lighting and label uncertainty instead of assuming lighting caused every error. → Plan controlled pair: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Errors per 100 photos in each lighting slice



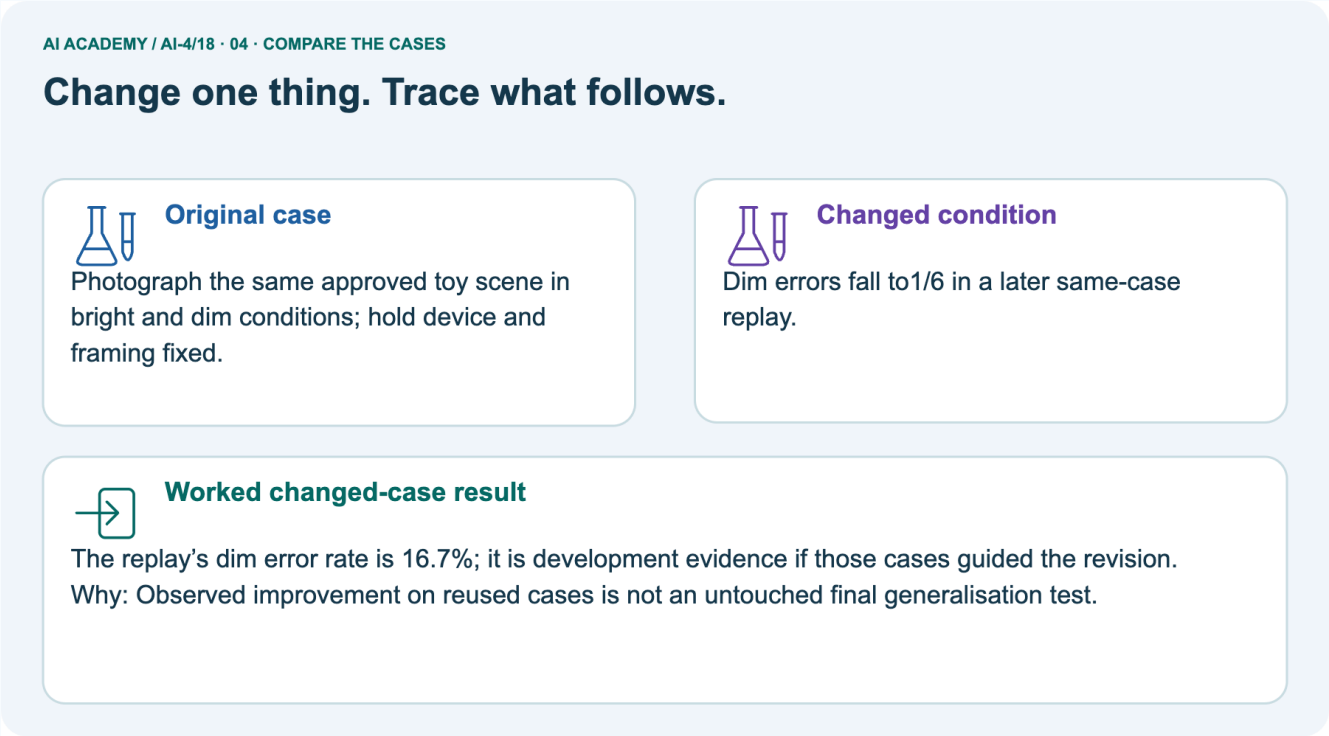
Follow the labelled picture with the matching explanation. Open the vector diagram

Errors per 100 photos in each lighting slice. Units: percent. bright: 12.5. dim: 50.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

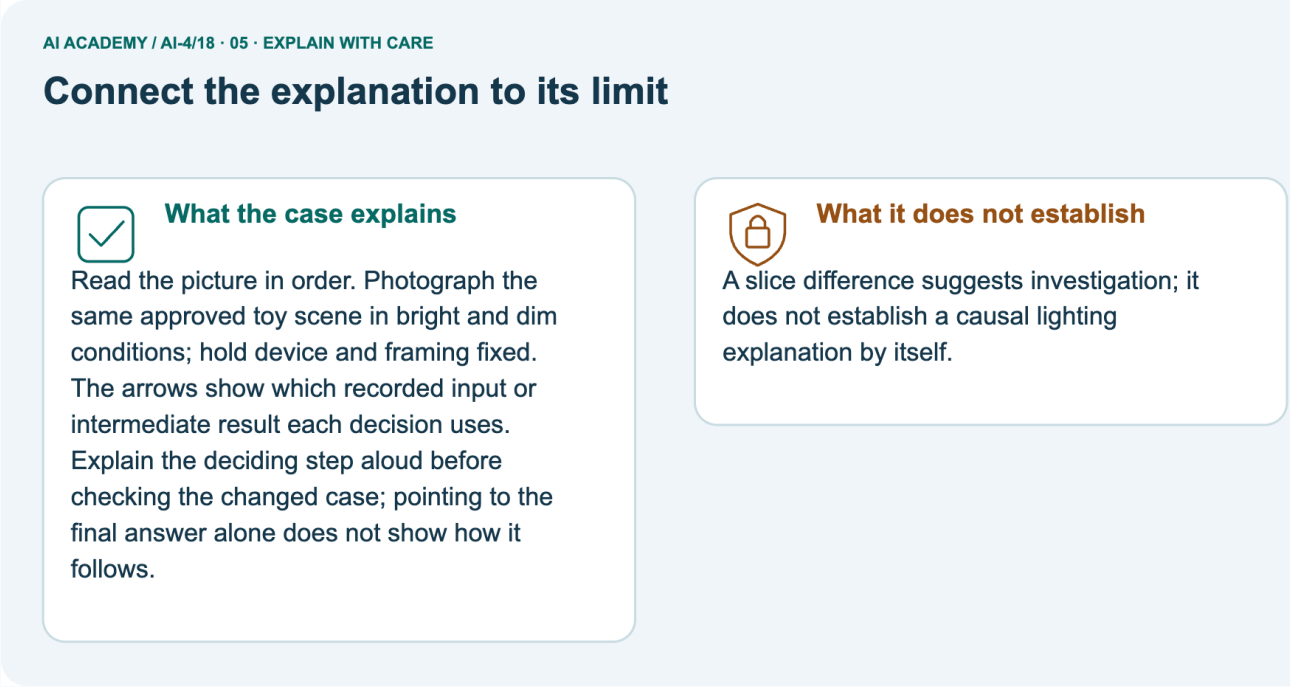
Original case: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Changed condition: Dim errors fall to1/6 in a later same-case replay.

Expected result: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Reason: Observed improvement on reused cases is not an untouched final generalisation test.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

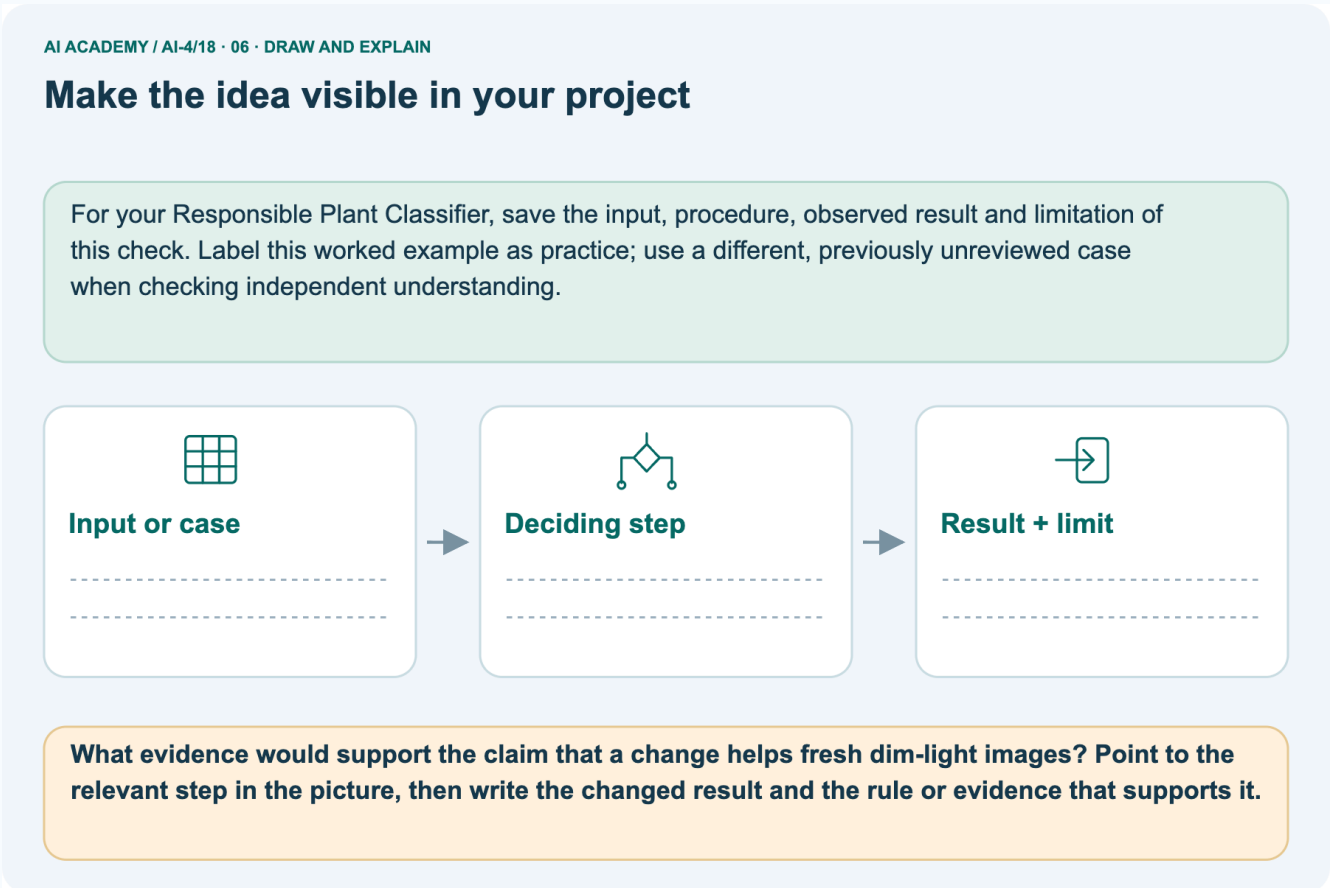
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.